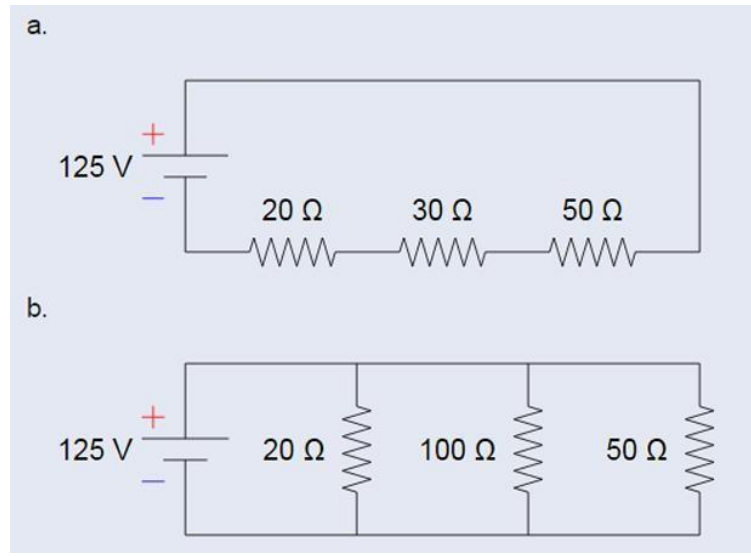


Part A



For each of these circuits:

1. Calculate the total resistance in ohms,

A = 100 Ohms

B = 12.5 Ohms

2. Calculate the total current from the power supply,

A = 1.25 A

B = 10 A

3. Calculate the current through each resistor

A = 1.25 A for all resistors

B = 20 Ohms: 6.25, 100 Ohms: 1.25, 50 Ohms: 2.5

4. Calculate the Voltage drop through each resistor

A = 20 Ohms: 25V, 30 Ohms: 37.5V, 50 Ohms: 62.5V

B = 125V for all resistors

5. Calculate the Power dissipated through each resistor (*use Watt's Law: Power = Voltage * Current*)

A = 20Ohms: 31.25W, 30Ohms: 46.875W, 50Ohms: 78.125W

B = 20Ohms: 781.25W 100Ohms: 156.25W 50Ohms: 312.5W